Vertex Algebras and Algebraic Curves — Notes

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These notes are based mainly on Frenkel & Ben-Zvi's *Vertex Algebras and Algebraic Curves*.

1 Chapter 1

Some notation for formal (power) series:

$$f \in \mathbb{C}[[z^{\pm}]] \iff f = \sum_{n\in\mathbb{Z}} f_n z^n,$$

$$f \in \mathbb{C}[z, z^{-1}] \iff f = \sum_{n_1 \le n \le n_2} f_n z^n,$$

$$f \in \mathbb{C}((z)) \iff f = \sum_{n\ge n_0} f_n z^n.$$

Note that the third is the ring of (formal) Laurent series, while the second is the ring of Laurent polynomials.

$$f \in \mathbb{C}((z))((w)) \iff f = \sum_{n\ge n_0} \sum_{m\ge m_n} f_{nm} z^m w^n.$$

Here m_n need not be bounded below, hence $\mathbb{C}((z))((w))$ and $\mathbb{C}((w))((z))$ are not isomorphic.

1.1 The definition of a VOA

Roughly speaking, a VOA is a Virasoro module together with a multiplication (OPE) structure. For instance, in a (p, p') minimal model, a single pair r, s gives a Virasoro module, and this module alone is not a VOA. Only after taking the OPE and generating all r, s does one obtain a VOA.

Here we quote the book directly.

Definition 1. A *vertex algebra* is a collection of the following data:

- *space of states*: a $\mathbb{Z}_+$-graded vector space $V = \bigoplus_{n=0}^{\infty} V_n$, with $\dim(V_n) < \infty$;
- *vacuum vector* $|0\rangle \in V_0$;
- *shift operator* $T : V \to V$ of degree one;
- *vertex operation*: a linear map $Y(\cdot, z) : V \to \operatorname{End} V[[z, z^{-1}]]$ taking each $A \in V_n$ to a field of conformal dimension n.

These data are subject to the following axioms:

- *(vacuum axiom)* $Y(|0\rangle, z) = \mathrm{Id}_V$. Furthermore, for any $A \in V$ we have $Y(A,z)|0\rangle \in A + zV[[z]]$ (i.e. $Y(A,z)|0\rangle$ has a well-defined value at $z=0$, which is equal to A).
- *(translation axiom)* For any $A \in V$, $[T, Y(A,z)] = \partial_z Y(A,z)$ and $T|0\rangle = 0$.
- *(locality axiom)* All fields $Y(A,z)$ are mutually local with each other.

The field condition: for $v \in V$, $A_n v = 0$ for n large enough.

Commutative operator algebra: all the $Y(A,z)$ commute with each other, equivalently each $Y(A,z)$ is regular at $z=0$. Equivalently, it is a commutative algebra with a derivation (possibly requiring some completeness conditions, otherwise $\exp(zT)$ cannot be defined).